Definitions

Def: An Airthmetic Progresion is a sequence $a_1, \ldots, a_n$ that satisifies the following invariant (It is defined invariantly)

$$\forall n \in \mathbf{Z}, \quad a_{n+2} - a_{n+1} = a_{n+1} - a_n$$

Theorems

From the invariant definition, we have the following properties

Theorem 1: For any arithmetic progression, the common difference is constant. That is, $\forall n \in \mathbf{N} \quad a_{n+1} - a_n = d$ where d is some constant.

Theorem 2: Let a_n be an arithmetic progression with common difference d. Then the a_i value of the sequence is given by $a_i = a_0 + i \cdot d$.

Thoerem 3: Let a_n be an arithmic progression with common difference d, then the sum of of values from $\{a_0, \ldots, a_k\}$ is given by $\sum_i a_i = k \cdot a_0 + \frac{k(k-1)}{2} d$.

Proofs

Theorem 1: For any arithmetic progression, the common difference is constant. That is, $\forall n \in \mathbf{N} \quad a_{n+1} - a_n = d$ where d is some constant

Proof of Thm 1:

Let a_n be an arithmetic progression. Let $d = a_1 - a_0$ be the common difference between the first two elements of the sequence. Then we must show the common difference is the same for every consequetive element of a_n. That is, we want to show

$$\forall n \in \mathbf{N} \quad a_{n+1} - a_n = d$$

Let $P(n)$ be the claim up to n. We will use induction.

Base Case: $P(0)$ then we must show $a_1 - a_0 = d$

This is true by the definition of d

Inductive Case: Suppose $P(k)$ is true for some partiuclar $k \geq 0$. We must show that $P(k+1)$ is also true. That is

$$a_{k+1+1} - a_{k+1} = d$$

Observe, we can use the invariant definition of airthmetic progression to write

$$a_{k+1+1} - a_{k+1} = a_{k+1} - a_k$$

Using the inductive hypothesis on the right hand side, gives the claim. This completes the inductive case $\square$.

Theorem 2: Let a_n be an arithmetic progression with common difference d. Then the a_i value of the sequence is given by $a_i = a_0 + i \cdot d$

Proof:

Let a_n be an arithmetic progression with common difference d. This means each consequtive elemnet in the seqence has distance $|d_k - d_{k+1}| = d$. We want to show a_i is given by $a_i = a_0 + i * d$.

We will prove the statement by induction. Let $P(n)$ be the statement with respect to n.

Base Case: $P(n = 0)$

We must show $a_0 = a_0 + 0 \cdot d$ is true. Oberve,

$$a_0 = a_0 + 0 \cdot d$$

Mul by zero

$$a_0 = a_0 + 0$$

Identity element (add by zero)

$$a_0 = a_0$$

This is true by reflexivity invariant of groups. Hence the base case is true.

Inductive Case: Suppose $P(k)$ is true for some $k \geq 0$. We must show that $P(k+1)$ is also true. That is

Assume: $P(k)\colon a_k = a_0 + k \cdot d$
Goal: $P(k+1)\colon a_{k+1} = a_0 + (k+1)d$

$$a_{k+1} = a_0 + (k+1)d$$

Distribute,

$$a_{k+1} = a_0 + d \cdot k + d$$

associativity

$$a_{k+1} = (a_0 + d \cdot k) + d$$

Use the inductive hypothesis

$$a_{k+1} = a_k + d$$

$$a_{k+1} - a_k = d$$

Use the invariant definition of arithmetic progressions

$$d = d$$

This is a true statement by reflexivity of additive commutative groups. This completes the base case. $\square$

Thoerem 3: Let a_n be an arithmic progression with common difference d, then the sum of of values from $\{a_0, \ldots, a_k\}$ is given by $\sum_i a_i = k \cdot a_0 + \frac{k(k-1)}{2}d$.